

LATENTSPRING: PHYSICS-INFORMED MOLECULAR FLOW MATCHING FROM ATOMIC COMPOSITION

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ABSTRACT

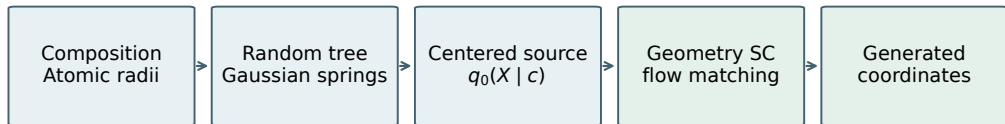
Generating molecular structures from atomic composition requires spatial organization to emerge without a supplied chemical graph. We introduce LatentSpring, a molecular flow-matching framework with a harmonic source that marginalizes over latent spatial dependence trees. Atom-dependent Gaussian edge displacements provide molecular length scales, while a matrix-tree ratio gives the normalized source density on centered coordinates. The resulting source integrates with an equivariant molecular flow without bond labels or an additional learned source network. On neutral-organic OMol25 compositions, matched continuations improve raw chemical-graph support over an isotropic Gaussian source by 5.86 percentage points on 12 development compositions, 8.83 points on 10 additional compositions, and 8.20 points on a subsequent frozen 24-composition test. A calibrated covariance-Gaussian comparison supports a contribution from mixture structure beyond the tested second-moment approximation. We further incorporate physical feedback during training: a paired parameter update reduces raw energy and force in both continuations under eSEN and independent GFN2-xTB readouts. Generation uses 128 primitive denoiser calls and requires no per-output energy selection or geometry optimization. A controlled molecular-rotor experiment also validates complete escorted-work correction of a specified local target. These results establish latent spatial dependence as a useful source design for composition-conditioned molecular flows and demonstrate a complementary route for incorporating physical feedback into the learned generator.

1 INTRODUCTION

Molecular generators must turn an unstructured coordinate distribution into a configuration compatible with its atoms. The source distribution determines part of this learning problem: independent Gaussian coordinates carry little molecular organization, while graph-conditioned harmonic priors encode relationships that must already be known. This distinction matters when the input specifies atomic composition and electronic conditions without a chemical graph, as in the bond-free OMol25 setting considered here.

LatentSpring introduces spatial dependence before learning the transport, while leaving the final chemical graph unspecified. We sample a latent tree over atoms, draw Gaussian displacements along its edges at atom-dependent scales, integrate them into coordinates, and remove translation. Marginalizing the tree defines a mixture of harmonic networks. The tree is an auxiliary source variable: the neural field receives coordinates and molecular conditions, and can generate chemical connectivities different from the sampled source tree. A matrix-tree expression makes the mixture density and its posterior-averaged elastic coupling explicit.

We integrate this source with equivariant flow matching and geometry self-conditioning. Controlled comparisons hold the backbone, initialization, training data, optimization and inference budget fixed while changing the source. The harmonic mixture improves raw graph support in both continuations and repeats its advantage on two subsequent composition panels. Comparing with both a narrower shell source and an approximate covariance-matched Gaussian distinguishes the useful source structure from simply imposing near-fixed distances or increasing second moments.



Source trees are latent dependence structures; no supplied chemical bonds

Figure 1: Implemented generation path. Atomic radii and species propensities define a random harmonic network used to draw coordinates. The field receives the coordinates and fixed conditions, not the sampled tree. Generation uses no per-output energy selection or structural relaxation. Physics-update variants incorporate physical feedback during training.

Physical feedback supplies a complementary training signal. Local force-escorted FIT samples define paired training targets, whose parameter update is transferred relative to a matched replay continuation. The final generator retains network-only inference, and a frozen new-composition test confirms energy and force improvements under two approximate potentials. Jarzynski-type complete-work weights provide a specified local energy-reweighting extension with explicit reference-density terms.

Our contributions are: (i) a normalized molecular harmonic-mixture source with latent spatial connectivity and no supplied bond graph; (ii) its integration with molecular flow matching and training-only physical feedback; and (iii) controlled source comparisons and new-composition, cross-potential evidence for raw generation quality. The construction builds on established tree-distribution, flow-matching and escorted-work principles. Our empirical target is raw coordinate generation; thermal-equilibrium sampling is a distinct claim with additional requirements.

2 RELATED WORK AND SCOPE

Sources and molecular flows. FlowMol3 combines joint topology/coordinate generation with self-conditioning, fake atoms and geometry distortion (Dunn & Koes, 2025). We reuse its equivariant vector-field implementation; self-conditioning is established prior art. ET-Flow constructs harmonic sources for graph-conditioned conformers (Hassan et al., 2024). Its molecular input graph and benchmark differ from our composition-conditioned coordinate task. Source-Guided Flow Matching studies guidance by changing sources under a frozen field (Wang et al., 2026); condition-dependent source learning is also established (Kim et al., 2026). Thus neither changing noise nor learning a source in general is our novelty claim.

Latent trees. Factored distributions over trees and tractable tree learning are classical (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Random-spanning-tree difference priors, including Gaussian mixtures and conjugate tree posteriors, have also been studied for imaging inverse problems (Everink, 2026). The molecular instantiation here uses three-dimensional isotropic edge displacements, an intrinsic centered coordinate measure, species-dependent scales, and no supplied molecular graph. Our experiments isolate the utility of this source structure through matched Gaussian, shell and covariance-Gaussian comparisons.

Target distribution and physics. OMol25 provides calculated structures and properties across electronic conditions (Levine et al., 2025); its empirical geometry distribution is not a specified-temperature canonical ensemble. Harmonic source energies encode spatial fluctuations, whereas an electronic potential describes different physics. Jarzynski’s relation and its single-molecule pulling applications connect nonequilibrium work to equilibrium free energies (Jarzynski, 1997; Hummer & Szabo, 2001). An arbitrary learned transport or higher graph-validity rate does not by itself supply the required work weights. We therefore separate our measured generation task from the optional energy-correction construction in Appendix E.

3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

Let $c = (z_1, \dots, z_N, Q, S)$ contain atomic numbers, total charge and spin multiplicity. Coordinates lie in $\mathcal{H}_N = \{X \in \mathbb{R}^{N \times 3} : \sum_i x_i = 0\}$, with intrinsic Lebesgue measure and dimension $3(N-1)$. For positive symmetric affinities a_{ij} , define

$$p(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(a)}, \quad \tau(a) = \sum_{T \text{ spanning tree}} \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

The selected fixed source uses $a_{ij} = u_i u_j$. We set $u_i = 0.05$ for H and F/Cl/Br/I, and otherwise $u_i = \max(1, \text{default valence}(z_i) - 1)$. This species heuristic discourages terminal atoms from becoming high-degree source hubs; it neither imposes nor predicts chemical bond orders. The fixed source uses species but not Q, S beyond validating their admissibility.

Gaussian edge laws and sampling. Let ℓ_{ij} be the sum of covalent radii and $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$, with $s = 0.2$ fixed throughout. The edge-displacement law is

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

The scale convention matches the covariance of the previously tested lognormal shell edges; it was not optimized on the confirmation outputs. Sample a tree using weighted Wilson sampling, draw its independent edge vectors, integrate along the tree, and subtract the coordinate mean. The resulting nominal source is invariant under rotations and permutations of identical species.

Proposition 1 (Intrinsic mixture density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

is normalized on $\mathcal{H}_N$. For a log affinity $\alpha_e = \log a_e$,

$$\partial_{\alpha_e} \log q_0(X | c) = \Pr(e \in T | X, c) - \Pr(e \in T | c). \quad (4)$$

Appendix B applies standard incidence-matrix and tree-ensemble identities. Two grounded Laplacian determinants evaluate the density in cubic time; the implementation uses positive log-space elimination for numerical stability. This is a density marginalized over trees, not conditional on the particular tree used to sample X_0 .

What physics is applied? For a fixed tree, define the dimensionless spring energy

$$\mathcal{U}_T(X) = \frac{1}{2} \sum_{(i,j) \in T} \frac{\|x_i - x_j\|^2}{\sigma_{ij}^2}. \quad (5)$$

The component is a normalized Gaussian proportional to $e^{-\mathcal{U}_T}$. The marginal $-\log q_0$ is a soft mixture of these spring energies, with the component normalization constants retained. Its coordinate score is

$$\nabla_{x_i} \log q_0(X | c) = - \sum_{j \neq i} \Pr((i, j) \in T | X, c) \frac{x_i - x_j}{\sigma_{ij}^2}. \quad (6)$$

Differentiating Equation (3) gives this posterior-averaged elastic coupling, which describes how the marginalized source organizes atomic positions. The springs have zero rest displacement; covalent radii set their fluctuation scale, not an excluded-volume guarantee or physical bond-equilibrium potential. The evaluated sampler draws this source; it does not use Equation (6) as an additional learned force or guidance term.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

Sample $X_0 \sim q_0(\cdot | c)$ and a reference X_1 from the training corpus. A type-preserving assignment and proper rotation couple the pair, followed by shared independent Haar rotation and within-label

permutation augmentation. This preserves the invariant source marginal while allowing dependence between paired endpoints (Appendix C). We use the linear path $X_t = (1 - t)X_0 + tX_1$ and target $V^* = X_1 - X_0$.

The FlowMol coordinate updater returns $D_\theta(X_t, t)$ and the research parameterization interprets $v_\theta = D_\theta - X_t$ as velocity. This explicitly trained displacement interpretation differs from a native endpoint head. A first pass yields $v^{(0)}$ and the endpoint estimate $\hat{X}_1 = X_t + (1 - t)v^{(0)}$. A second pass recycles that estimate through the existing geometry self-conditioning modules to obtain $v^{(1)}$. Both passes remain differentiable, with objective

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\|v^{(0)} - V^*\|^2 + \|v^{(1)} - V^*\|^2 \right]. \quad (7)$$

Atomic categories and the declared charge interface remain fixed; edge inputs carry the constant no-bond interface. No bond-classification loss is introduced. The sampled tree is not supplied to the vector field. Self-conditioning and the endpoint-estimation identity are existing techniques shared by all primary arms.

Generation integrates $v^{(1)}$ with 32 midpoint steps and two passes per field call, totaling 128 primitive denoiser evaluations per attempted sample, followed by 0.025 Å centered Gaussian noise. We retain this numerical sampler exactly for comparisons. Source normalization alone does not give the finite integrator plus noise a qualified final density. Source-comparison training uses FM only; existing force/energy training interfaces are not evidence for a thermal law, and the independent-Gaussian velocity-to-score proxy is invalid for this source/coupling.

4.1 PHYSICAL FEEDBACK THROUGH PAIRED TRAINING TARGETS

A frozen harmonic generator supplies FIT anchors A on compositions excluded from evaluation. For graph-supported anchors, a bounded force displacement defines local escorted samples $Y = X + \delta_A$, with $X \sim q_A = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. The local scale and displacement are fixed from the anchor force before drawing particles (Appendix F). We train a physical-target continuation and a replay continuation from the same weights and matched training draws, then transfer their parameter difference:

$$\theta_* = \theta_0 + (\theta_{\text{physical}} - \theta_{\text{replay}}). \quad (8)$$

This applies established parameter arithmetic (Ilharco et al., 2023) with coefficient one and fixed model buffers. The update changes the learned generator, so inference requires no force calls or structural relaxation.

Complete escorted-work weighting (Vaikuntanathan & Jarzynski, 2008) defines a local physical extension. On its specified structural support, let

$$W_A = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)]. \quad (9)$$

The full weight includes the reference-density correction for the escort, rather than only an energy difference. Its expectation gives the normalizer of the Gaussian-restrained local energy tilt. We evaluate both uniform and complete-work weights; the simpler force update is the prespecified candidate for the fresh cross-potential test. Appendix F provides the derivation, finite-particle implementation and matched training controls.

5 EXPERIMENTAL DESIGN

Task and training. We evaluate coordinate generation for neutral-singlet organic compositions, containing C and H and elements from H/B/C/N/O/F/Si/P/S/Cl/Br/I. The implementation supports 200 atoms; experiments are limited to 8–40. Two continuations share one pretrained warm checkpoint. Each uses 3,000 distinct OMol25 training rows and 3,000 optimizer updates, with identical rows and initial weights across source arms within that continuation. The source is fixed and adds no fitted parameters. The vector-field parameter count is 5,904,369. Backbone learning rate is $2 \cdot 10^{-5}$; activated self-conditioning modules use $3 \cdot 10^{-4}$. Two differentiable passes are used for every primary source arm.

The training subset is deliberately narrower than the full processed corpus: neutral-singlet organic references must pass the unchanged graph assay and admit the declared bounded-edge, degree-capped

Table 1: Chemical-graph support per attempted output. All primary arms share the geometry-SC backbone and 128 primitive denoiser calls. The additional panel uses frozen models selected after the development comparison.

Panel	Continuation	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

geometric scaffold used in an earlier training selection. The source-SC models reuse these coordinates but ignore the stored scaffold. All arms share this selection. It is derived geometric selection, not supplied bond supervision, and limits claims about unrestricted OMol25. The data selection and training parameters are retained in the frozen protocols.

Panels and separation. Original reference qualification identified 31 accepted compositions among 78 metadata-eligible candidates. An earlier 21-composition panel used up to 8 per size bin; the present 12-composition development panel is a fixed subset of that panel. After comparing all three source arms, we selected harmonic sources jointly across the two continuations and froze both harmonic and Gaussian models. We then evaluated *all* remaining 10 qualified compositions. They happen to have 17–28 atoms; no generator output was used to filter them. Exact count-vector checks find no matching composition in 3,902,107 processed training rows or 39,415 validation rows. References qualify the assay and do not initialize the generator or enter fitting. Separation from these corpora does not certify unrecorded pretraining or all trajectory relations. The additional panel reuses the same trained models, not two further independent training replicates.

Metrics and uncertainty. The first two panels use 64 samples per composition; the fresh cross-potential panel uses 32. Geometry support requires contact connectivity and no short-distance overlap using 1.25 and 0.6 times summed covalent radii. Graph support additionally requires successful chemical-graph perception. This assay does not certify quantum stability, the complete electronic state or synthetic accessibility. Failed outputs and validator exceptions stay in the denominator. We also record distinct perceived connectivity within each composition and actual generation time. Paired bootstrap intervals condition on the fitted models and fixed compositions; a composition bootstrap separately summarizes heterogeneity. Neither is a broad population-generalization certificate.

6 RESULTS

Harmonic trees improve graph support over isotropic Gaussian sources by 5.86 percentage points on the development panel, with conditional paired 95% interval [2.67, 8.98] and descriptive composition interval [0.52, 12.17]. Both continuation signs are positive, although the first gain is substantially larger. The narrow shell source fails its prespecified superiority gate and is worse than the harmonic tree by 4.23 points, with paired interval [-7.29, -1.11].

On the additional 10 compositions, harmonic sources improve graph support from 52.73% to 61.56%, or 8.83 points, with paired interval [5.08, 12.50] and descriptive composition interval [4.84, 12.66]. Geometry support improves by 9.69 points, with paired interval [6.33, 12.89]. All graph-supported samples have distinct perceived connectivity within their composition in this panel. This is an assay-specific diversity readout; structural stereochemical novelty is not inferred from it. No validator exception occurs. Harmonic/Gaussian generation times are 175.73/175.43 seconds and 178.20/177.74 seconds for 640 samples. These measurements do not establish a speed advantage. All 2,560 additional-panel source and structural records replay.

Control for source covariance. A single Gaussian source uses the existing 128-tree estimate of the harmonic mixture covariance, symmetrized over identical species. Calibration against an independent 8,192-tree estimate gives 1.85–6.25% relative Frobenius error across all 22 compositions; this control is explicitly approximate. Its source settings were inherited from the earlier control

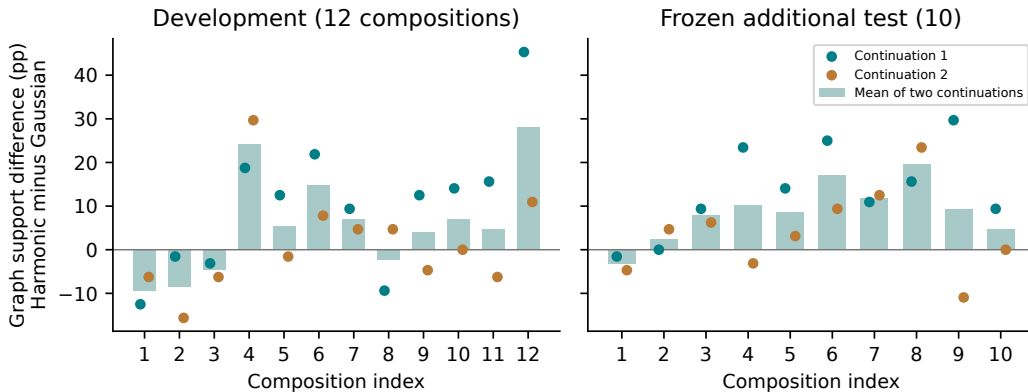


Figure 2: Harmonic-source graph-support gains vary across compositions and continuations. Points show individual continuation differences; bars average them. Positive aggregate gains do not imply improvement for every composition. The additional panel uses the same two models as development.

Table 2: Covariance-Gaussian control at matched continuation and inference budgets. Its covariance is a finite-tree approximation, not an exact identity.

Panel	Continuation	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

implementation. It was added after the initial additional-panel result, with the same data, optimizer, network and sampling budgets; neither harmonic model was changed.

Harmonic sources improve graph support by 11.65 points on development (paired 95% [8.59,14.78]; descriptive composition [8.07,15.43]) and 15.39 points on the additional panel (paired 95% [11.64,19.06]; composition [11.95,18.98]). All four point differences are positive. The covariance Gaussian is also worse than the isotropic Gaussian in both panels. Thus simple covariance replacement does not explain the observed harmonic-source advantage in these continuations. Finite covariance-estimation error and model optimization remain limitations of an exact higher-moment attribution. All 2,816 new control outputs replay, and parameter counts, initial feedback weights, data rows and training recipes match the frozen references. This is a new control on examined panels, not another untouched test.

The additional panel also has an eSEN readout and a task-adapted original-endpoint reference (Appendix A). Their differing continuation effects and training histories limit the corresponding comparison claims.

6.1 FRESH COMPOSITION AND CROSS-POTENTIAL CONFIRMATION

After the earlier source and physics studies, we freeze the Gaussian model, LatentSpring’s harmonic model, and both fixed-coefficient physics-update variants. The cheaper unweighted-force update is designated the primary physical candidate before querying new outcomes. We sample a new pool of 256 distinct compositions from the original development partition, excluding every old official candidate composition. Reference qualification accepts 121; fixed metadata ranks select 8 per 17–20, 21–24 and 25–28 atom bin, yielding 24 new compositions. All selected compositions are absent from both verified processed corpora. The reference coordinates do not enter generation or fitting. Every model generates 32 samples per composition in each of the same two existing continuations: 6,144 new outputs, with no additional model fitting or parameter selection. Inference uses the network alone: no per-output energy ranking or coordinate optimization is applied.

Table 3: Fresh 24-composition graph support, 768 attempts per continuation. All rates describe raw outputs before any structural relaxation. All methods use 128 primitive denoiser calls per attempted output.

Model	Continuation 1	Continuation 2	Pooled rate
Gaussian reference	332	352	44.53%
LatentSpring harmonic	435	375	52.73%
With force update	439	382	53.45%
With complete-work update	426	383	52.67%

Table 4: Primary force-update change versus the frozen harmonic model on paired common-graph support. Both continuation signs improve for both readouts under both potentials. Intervals are conditional paired 95% intervals.

Potential	Energy change (eV/atom)	Force-RMS change (eV/Å)
eSEN	−0.01308 [−0.01805, −0.00817]	−0.34793 [−0.41498, −0.28478]
GFN2	−0.01168 [−0.01616, −0.00724]	−0.35620 [−0.42983, −0.28714]

The harmonic source improves graph support over Gaussian by 8.20 percentage points, with conditional paired 95% interval [5.08, 11.33]. Both continuation signs are positive. The primary force update has an observed graph-rate difference of +0.72 points versus harmonic, with interval [−0.98, 2.34]. This is compatible with unchanged validity and excludes a decline exceeding two points under this conditional interval; it is not evidence of a statistically significant validity increase or a population-wide noninferiority guarantee. All accepted graphs have distinct perceived connectivity within each composition in these draws.

Independent physical readout. We evaluate every generated coordinate set with eSEN and with gas-phase GFN2-xTB single points and analytic gradients (Bannwarth et al., 2019). GFN2 is not used for training or method selection. It provides a different approximate physical model, not DFT ground truth. We retain all failed calculations: 4 of 6,144 generated-geometry GFN2 calculations fail, all outside graph support. Every graph-supported output succeeds, and all original-reference inversion checks pass. Total physical cost is 12,384 raw eSEN queries and 6,240 GFN2 attempts, including reference checks. The audited eSEN readout uses $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$; force RMS is the root mean squared atom-force norm. GFN2 inputs are fixed at their saved generated coordinates. The xTB command uses `--grad`, never `--opt`; gradients are readouts and do not update coordinates. Physical teacher displacements occur only in the earlier training procedure, on separate FIT samples.

All 24 composition cells in each continuation have comparable graph-supported pairs for the primary contrast. The prespecified cross-potential confirmation gate passes, including success coverage and the observed two-point validity criterion. All-attempt eSEN and common-success GFN2 comparisons agree in direction; selected-support results remain distinct from unconditional or equilibrium expectations. Source-only harmonic-versus-Gaussian energy improvements are pooled positive but heterogeneous across continuations, so we do not describe them as repeated physical-quality gains in both models.

Complete-work versus force-update energy differences remain unresolved: −0.00413 eV/atom under eSEN, interval [−0.00903, 0.00078], and −0.00363 under GFN2, interval [−0.00792, 0.00075], on common graph support. Force and graph intervals also span zero. Thus the fresh result supports the practical physical feedback update, while not establishing an extra advantage of Jarzynski weighting. The previously defined local work identity remains mathematically useful, but these results do not imply globally Boltzmann-distributed generator outputs.

6.2 AN INDEPENDENTLY ADAPTED DIFFUSION BASELINE

We additionally adapt the official EGNN/EDM backbone (Hoogeboom et al., 2022) to the same composition-conditioned task. Positions follow its variance-preserving diffusion schedule, while

Table 5: Raw graph support of the independently adapted conditional EDM, 768 attempts per continuation. The harmonic and force-update models in Table 3 reach 52.73% and 53.45%, respectively, at 128 calls.

Sampler	Continuation 1	Continuation 2	Pooled rate
EDM, 128 calls	103	100	13.22%
EDM, 1,001 calls	93	106	12.96%

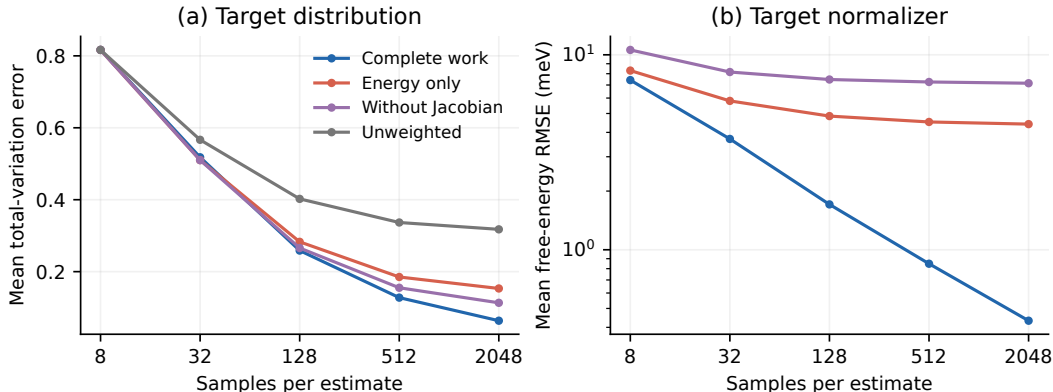


Figure 3: Complete work corrects the nonequilibrium sampling bias on the specified torsional targets. Curves average all three molecules and both nonidentity escorts; each case has 128 repetitions. Energy-only and incomplete-work controls retain bias as sampling increases. Unweighted endpoints have no free-energy estimator. These are constrained molecular mechanism tests, not whole-generator equilibrium results.

clean atom categories are conditions during both training and sampling. The GEOM-pretrained checkpoint receives 30,000 updates on the same 3,000 OMol25 FIT rows used by each source continuation, with its original 10^{-4} learning rate. We evaluate both 128 denoiser calls and the full 1,001-call sampler, including the final observation step, on all 24 fresh compositions.

This complements the matched source ablations by testing a different pretrained backbone and generative objective. The backbones and pretraining histories differ: EDM has 2.38 million parameters and one denoiser pass per update, versus 5.90 million and two for the source-SC models. Its longer continuation strengthens task adaptation, but does not match total pretraining resources or certify separation from its GEOM pretraining data. The table reports this conditional adaptation protocol, not EDM’s published joint-generation benchmark. Equal call counts also differ in FLOPs; generation and training times are recorded separately.

6.3 TESTING COMPLETE WORK ON MOLECULAR TORSIONAL TARGETS

We isolate the role of complete work using three geometry-qualified methyl rotors from FIT molecules, selected in fixed data order before querying their energies. Each rotor defines a constrained one-dimensional molecular curve. A periodic interpolant of 512 GFN2 energies supplies the potential $U(\phi)$; the target is $\rho(\phi) \propto q_0(\phi) \exp[-(U(\phi) - U(0))/kT]$, where q_0 is a normalized von Mises reference of concentration one and kT corresponds to 300 K.

Starting from $\phi \sim q_0$, use escorts $\psi_a(\phi) = \phi + a \sin \phi$ for $a \in \{-0.5, 0, 0.5\}$. Their positive Jacobian is $J_a = 1 + a \cos \phi$. The complete generalized work includes both reference densities and $-kT \log J_a$ (Appendix G). We compare complete work, energy-only weights, weights omitting the Jacobian, and unweighted endpoints, with shared draws and 128 repetitions at each of 8, 32, 128, 512 and 2,048 samples.

At 2,048 samples, complete work has lower mean total-variation error than both incomplete-weight controls in all six nonidentity cases. Across these cases, mean errors are 0.0636 for complete work, 0.1532 for energy-only weights and 0.1134 without the Jacobian, a 58.5% reduction relative to energy-

only weighting. Complete-work free-energy RMSE ranges from 0.158 to 0.734 meV. Independent density-ratio quadrature also recovers the specified target normalizer in all nine molecule/escort cases. The identity-escort control makes the three weighted methods coincide. This validates the implemented work correction on these constrained targets; its additional benefit after neural distillation is assessed separately from the exact identity.

7 DISCUSSION AND LIMITATIONS

LatentSpring shows that a fixed, structured source can improve a molecular flow without a supplied graph or an additional learned source head. Marginalizing spatial dependence trees provides a tractable probabilistic construction, and the source comparisons support its practical value across the reported composition panels. Training-only physical feedback further improves raw energy and force readouts while preserving the observed graph-support rate in the fresh test.

The evaluated domain is a selected neutral-organic subset of OMol25; models share a warm start, and the tests reuse two fitted continuations. Conditional intervals quantify sampling uncertainty on these compositions, rather than uncertainty over all possible retrained models. The measured advantage at this continuation budget does not establish an asymptotic advantage after converged large-scale training. Graph support is a structural assay, and the two energy models are approximate: these readouts do not imply proximity to converged quantum minima, synthetic accessibility, or a canonical energy distribution. The local escorted-work identity in Appendix F and global connection in Appendix E specify their respective targets and assumptions. The current evidence supports the source and physical-feedback construction for raw generation, with broader chemical coverage and equilibrium sampling as separate directions for development.

REPRODUCIBILITY STATEMENT

Frozen protocols specify training and generation seeds, data selection, model checkpoints, source distributions and physical-evaluation settings. All attempted outputs, including structural and oracle failures, are retained. Artifact audits check source sampling, coordinate and condition identities, structural readouts, work weights and parameter-transfer arithmetic. Model-training and inference budgets are reported separately from physical teacher and evaluation costs. Implementation tests cover source normalization, symmetry, density gradients, escort Jacobians and work identities. The code and artifact manifests accompany the research record; source data and pretrained backbones retain their original licenses and access requirements.

AI USE STATEMENT

AI tools assisted research exploration, literature checks, mathematical derivation, implementation, testing, experiment orchestration, analysis and drafting. Experimental protocols, implementation checks and underlying records are retained for verification.

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A EARLIER PHYSICAL AND ENDPOINT-REFERENCE READOUTS

Energy-model readout of the frozen generators. We score every output on the ten-composition panel and each original reference with the same conservative eSEN checkpoint at its original charge and spin. To retain the parity convention of the earlier oracle audit, use $E_+(X) = [E(X) +$

Table 6: Harmonic minus isotropic Gaussian physical readouts on the frozen additional panel. Lower is better for these surrogate-quality measures. Intervals are conditional paired 95% intervals.

Readout	Support	Difference	Interval
Energy (eV/atom)	All attempts	-0.0406	$[-0.0579, -0.0232]$
	Common graph	-0.0340	$[-0.0692, -0.0003]$
Force RMS (eV/Å)	All attempts	-0.4635	$[-0.6782, -0.2512]$
	Common graph	-0.5389	$[-0.8950, -0.2089]$

$E(-X)/2$ and $F_+(X) = [F(X) - F(-X)]/2$. The model measures are an energy proxy and force consistency, not independent quantum validation. Differences are paired within composition before averaging; energies per atom also subtract that composition’s reference energy, which cancels in the paired difference. There are 5,160 raw energy/force queries, including both inversions of the references in each continuation’s readout. All stored inputs, outputs and derived quantities replay in the audit.

Force RMS means the square root of the per-atom mean squared force norm, then averaged over samples. The common-graph statistic uses only paired outputs accepted by both methods; cell counts range from 3 to 35. It is a selected-support estimand and is not an unbiased equilibrium expectation. All-attempt statistics include outputs rejected by the structural assay and may reflect potential-model extrapolation. The two readouts therefore answer different questions and are both retained. Their positive direction strengthens the quality interpretation of the source result, but lower energy or force magnitude does not establish a Boltzmann distribution or remove the need for independent physical validation.

Continuation effects are heterogeneous. All-attempt energy differences are -0.08049 and -0.00070 eV/atom; force-RMS differences are -1.21996 and $+0.29302$ eV/Å. Thus the pooled force advantage does not replicate in sign across both continuations, and the second energy effect is small. The conditional intervals do not quantify uncertainty over independently retrained models.

Original endpoint-checkpoint reference. We also evaluate the original 30,000-step OMol25 checkpoint using its endpoint semantics, 127 Euler updates and first-step bootstrap/history conditioning, for 128 actual denoiser calls. Current composition and the neutral/no-bond interface are clamped to define the same coordinate-generation task. Native predicted history gives 106/640 and 113/640 graph passes; a variant with fixed history categories and a centered first bootstrap gives 116/640 and 120/640. The two streams use the same original checkpoint, not two training replicates. All 2,560 outputs replay. The latter changes bootstrap centering as well as categorical history and is not a pure categorical-feedback ablation. This is a task-adapted reference, not the published joint-generation benchmark. The source-SC models have extra shared adaptation and 3,000 specialized updates; these differences cannot be attributed solely to source choice. The matched Gaussian and covariance-Gaussian continuations remain the primary source controls.

B NORMALIZATION AND AFFINITY-GRADIENT PROOF

For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be an oriented incidence matrix and $U \in \mathbb{R}^{N \times (N-1)}$ an orthonormal basis of the centered subspace. Because $B_T \mathbf{1} = 0$, $(B_T U)(B_T U)^\top = B_T B_T^\top$. Cauchy–Binet expresses $\det(B_T B_T^\top)$ as the sum of squares of its N maximal minors. Each minor of a tree incidence matrix is ± 1 , so the determinant is N . The absolute Jacobian in one spatial dimension is therefore $\sqrt{N}$, and in three dimensions it is $N^{3/2}$. Consequently the component density is $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$ and integrates to one. Averaging with Equation (1) yields Equation (3) and preserves normalization. Differentiating the log of each finite positive tree partition with respect to α_e gives its edge-inclusion probability, proving Equation (4). These are applications of standard incidence and tree-ensemble identities.

C SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Suppose q_0 is G -invariant, $X \sim q_0$, and a pairing algorithm selects $g(X, Y) \in G$ using a target Y . If H is an independent Haar draw, then conditional on the orbit of X and on the selected g , HgX has the uniform orbit distribution of HX , by Haar invariance. Integrating over the unchanged source-orbit distribution recovers q_0 . This argument permits a non-Gaussian source and explains the shared augmentation used in the pairing. It does not imply independence of source and target after pairing or justify the independent-Gaussian velocity-to-score formula.

D COVARIANCE CONTROLS

For $R = \ell \exp(sZ - s^2/2)$ with $Z \sim N(0, 1)$, $\mathbb{E}R^2 = \ell^2 \exp(s^2)$. A uniform direction has second moment $I/3$, so each edge has covariance $vI = \ell^2 \exp(s^2)I/3$. Let A_T map edge coordinates to centered node coordinates. Independence gives $C_T = A_T \text{diag}(v_e) A_T^\top$. Replacing the edge vectors by independent Gaussians of covariance $v_e I$ leaves C_T unchanged and, since all component means are zero, leaves $\mathbb{E}_T C_T$ unchanged. The single-Gaussian control uses a finite, composition-seeded average of these matrices. Averaging this matrix over within-species permutations preserves positive definiteness on the centered subspace and yields species invariance. The Gaussian is exactly normalized for its chosen covariance; equality to the infinite tree-mixture covariance remains approximate.

Additional source-adaptation prior art. Source-Guided Flow Matching already obtains guidance by changing the source under a frozen vector field, including source-space importance and MCMC sampling (Wang et al., 2026). Conditional source learning within flow matching also appears in Kim et al. (2026). Random-tree continuous difference priors have imaging applications (Everink, 2026), and molecular diffusion fine-tuning with physical rewards appears in Zhou et al. (2025). A positive pilot would therefore establish only an internal molecular mechanism result. Comparative utility and cost would still be needed to substantiate the specific structured source adaptation contribution.

E CONNECTION TO JARZYNSKI AND ESCORTED TRANSPORT

The motivating connection to nonequilibrium force measurements is retained, but is separate from the evaluated validity experiment. Jarzynski relates exponential work averages to equilibrium free-energy differences under a qualified nonequilibrium protocol; Hummer–Szabo reconstruction applies this principle to single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Artificial escort flows generalize the transport protocol while retaining the appropriate work accounting (Vaikuntanathan & Jarzynski, 2008). None of these identities is a new result of this paper.

A change-of-variables version clarifies what our normalized source could supply. Fix a composition and intrinsic coordinate measure. Let $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$ be a target with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$, for example a declared confined molecular ensemble. Normalizability must not be assumed for dissociating fragments in unbounded space. Let $q_0 > 0$ be the normalized source in Equation (3), and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable bijection with nonzero intrinsic Jacobian. For $\Phi(x_0) \in \Omega$, define the generalized work

$$W_\Phi(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (10)$$

and set $W_\Phi = +\infty$ otherwise. Then

$$\mathbb{E}_{q_0}[e^{-\beta W_\Phi}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (11)$$

$$= Z_1. \quad (12)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in the declared measure, yielding a Jarzynski-type normalizer identity. The Jacobian term accounts for compression by the escort map; the work cannot be replaced by an endpoint energy difference alone. Weighted observables require the same complete weight. Even a correct identity need not give an efficient finite-sample estimator when the weights have poor overlap.

For a sufficiently regular exact ODE flow, the intrinsic log determinant is the integral of the intrinsic velocity divergence. Our measured midpoint sampler followed by terminal Gaussian noise is not that exact diffeomorphism. A valid implementation would need either its own qualified map/Jacobian or a specified forward/reverse stochastic path law that includes the noise and every correction. We have not performed this global reweighting for the source-SC results. A separate local anchored work teacher is defined and tested in Appendix F. No global work-variance, free-energy, effective-sample-size or Boltzmann-distribution claim is inferred from their graph or geometry pass rates.

F LOCAL FORCE-ESCORTED WORK DISTILLATION

This separate experiment brings energy/force feedback into training while keeping the positive source-comparison checkpoints frozen. Energy-weighted flow matching and weighted physical-sample training already exist (Dern et al., 2025; Midgley et al., 2023); escorted work identities are also established (Vaikuntanathan & Jarzynski, 2008). Our local construction is an application of these principles, not a new global Boltzmann-sampling theorem.

A specified local reference and target. For a generated FIT anchor A with accepted perceived graph G_A , define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. At $kT = 0.0258519998$ eV (300 K), remove net force and set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (13)$$

A vanishing force selects the maximal width. These quantities are fixed before drawing $X \sim q_A$. The escort $Y = X + \delta_A$ is a translation on the centered subspace and has determinant one. The local unnormalized target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y), \quad (14)$$

where the support requires the same labelled perceived bond orders, formal charges and radical counts, as well as the existing connected/nonoverlap assay. These are derived structural checks, not supplied bond labels or a bond-classification loss. This target includes a Gaussian restraint around A and does not equal the unrestricted molecular Boltzmann law.

Complete local work. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus hard support, the generalized quench work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (15)$$

on valid support, and $+\infty$ otherwise. Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. The two reference-density terms correct for the escort’s proposal shift. Energy difference alone would omit them. Two analytic tests verify constant work for a constant-force escort and the normalizer/weighted mean for a quadratic energy. These tests validate the local arithmetic, not a global molecular sampler.

Training and controls. Each continuation selects eight distinct 17–28-atom TRAIN compositions by a fixed metadata hash and generates 32 new FIT anchors per composition. They are excluded from the evaluated compositions. Invalid anchors are recorded rather than replenished. Each valid anchor produces eight escorted particles. Particles outside its declared support receive zero weight. The work arm samples training endpoints using self-normalized $\exp(-W_A/kT)$ weights; the unweighted escort arm samples the same valid particles uniformly. A replay arm uses the retained raw anchors, and a frozen model supplies the reference. All-zero-particle anchors are recorded and excluded identically from all student pools.

Each student alternates 500 original-reference FM updates with 500 generated-FIT endpoint updates, sharing initialization, anchor draws, optimization and sampling budgets. Physical preparation costs differ: replay needs no force queries, the unweighted escort needs anchor forces, and complete work additionally needs energies at valid proposed endpoints. Actual shared research cost and these method requirements are reported separately. No oracle is queried by the final generator. Physics

Table 7: Direct physics-distillation graph support per 640 attempts. The frozen reference is not refitted; other arms have identical additional update budgets.

	Continuation	Frozen	Replay	Unweighted escort	Complete work
1		418	340	330	344
2		377	345	371	362

enters through training targets. The generator retains its legacy constant temperature input; this is not a model trained to predict different temperatures. Finite-particle normalization biases the teacher relative to the exact normalized local target. Its ESS refers only to eight local particles, not global generator outputs. Original-reference mixing and unknown chemical-isomer weights further prevent a global equilibrium claim, even if the local identity is exact.

Direct-distillation outcome. Both continuations finish all three 1,000-update student fits. The 512 generated FIT anchors yield 351 graph-supported anchors, of which 349 have at least one supported escorted particle. All original draws and unsupported particles remain in the records. Mean local particle ESS by composition is about 1.27–1.87 out of 8, indicating substantial weight concentration and limited finite-particle resolution of the local target.

On paired common-graph support, complete work versus the unweighted escort reduces mean energy by 0.00758 eV/atom (conditional paired 95% interval $[-0.01424, -0.00113]$ for the signed difference) and force RMS by 0.16174 eV/Å (interval $[-0.27660, -0.05590]$). Both continuation signs favor complete work. All-output readouts agree in direction. Thus an incremental physical-quality signal is present relative to the matched escort student.

However, all direct continuations harm the frozen model. Complete work has graph support 55.16%, versus 62.11% for frozen, and common-supported energy is *higher* by 0.10587 eV/atom (interval $[0.09009, 0.12168]$). The prespecified adoption gate therefore fails. The matched physics increment does not establish an overall generator improvement. The replay control shows that plain continuation on these mixed targets already produces substantial regression; it does not uniquely identify the cause of that regression.

All 512 FIT source draws, 5,120 evaluation outputs, local work values, support masks, selections, checkpoints and physical readouts are audited. Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at supported particles. An unweighted escort could omit the particle-energy queries; replay could omit all physical queries. Evaluating the four generated-output arms uses another 10,240 queries, reported separately from inference cost. No global Boltzmann, output-ESS, work-law novelty or successful direct-distillation claim follows from this result.

Separating the paired update from continuation regression. The failed direct continuation motivates one subsequent fixed-coefficient parameter-arithmetic test. Parameter arithmetic is established (Ilharco et al., 2023); we make no claim to invent it or to subtract functional drift exactly. For the same initialization and paired training draws, construct

$$\theta_{\text{new}} = \theta_{\text{frozen}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (16)$$

for both unweighted-escort and complete-work students. The coefficient is one; there is no coefficient, layer-selection or extra-training sweep. Only parameter tensors change, while all buffers and the source distribution remain fixed. All four reconstructed checkpoints are checked tensor by tensor and restore.

Both continuation signs improve for energy and force in both transferred variants. The work-difference variant reduces force RMS by 0.38403 eV/Å on common graph support (interval $[-0.48124, -0.28932]$); all-attempt energy and force readouts also improve. Its observed graph-rate difference is -0.55 percentage points, with interval $[-2.58, 1.56]$. The frozen gate used a two-point tolerance on the *observed pooled rate*; this is not a formal statistical noninferiority certificate, and a larger loss remains compatible with the interval.

The additional effect of complete work over the transferred unweighted escort is not established: common-supported energy difference is -0.00243 eV/atom, with interval $[-0.00756, 0.00264]$,

Table 8: The single parameter-arithmetic test preserves most observed graph support while improving the eSEN readout. Energy differences are relative to frozen on paired common-graph support, with conditional paired 95% intervals. This test was designed after the direct-continuation result on the same panel.

Model	Graph passes (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Escort difference	407/640; 387/640	-0.01554 [-0.02084, -0.01031]
Work difference	406/640; 382/640	-0.01604 [-0.02188, -0.01013]

while all-attempt energy difference is +0.00022 with interval $[-0.00557, 0.00649]$. Force and graph intervals also span zero. Consequently the practical physical-quality signal does not require an asserted Jarzynski-specific advantage. The unweighted variant would require only 702 anchor queries for teacher preparation, versus 6,232 for the complete-work variant. Both use the same single-network 128-call inference budget.

This exploratory intervention passes the frozen-baseline physical-improvement gate for work transfer, but fails the separate extra-work-over-escort gate. It adds 2,560 generated outputs and 5,120 physical queries. No new optimizer steps are used in the transfer itself. Existing panel outcomes informed the choice of intervention, motivating the subsequent fresh-composition and independent-potential confirmation in Section 6.1. These results support investigating the simpler physics update while retaining complete work as a defined and tested comparison; they do not establish global Boltzmann sampling or new task-arithmetic theory.

G MOLECULAR ROTOR WORK PROTOCOL

The three FIT molecules are $\text{C}=\text{C}(\text{N})\text{N}(\text{C})\text{C}(\text{N})=\text{O}$, $\text{CCOC}(\text{C})(\text{C})\text{N}$, and $\text{CC}(=\text{O})\text{NCC}=\text{O}$. Selection examines the first five rows of an existing training subset, accepts the first three distinct compositions with a methyl rotor preserving the labeled perceived graph on a 64-angle grid, and queries no energy during selection. Only methyl hydrogens rotate around the carbon-anchor axis; the remaining internal coordinates stay fixed. Overall translation is removed. Density is defined with respect to the periodic angle measure $d\phi$ on $[-\pi, \pi)$, not full Cartesian molecular volume.

For $q_0(\phi) = \exp(\cos \phi)/(2\pi I_0(1))$, define reference Hamiltonian $H_0 = -kT \log q_0$ and target Hamiltonian $H_1 = H_0 + U - U(0)$. The escort has $|a| < 1$, so it is a periodic diffeomorphism. Its complete work is

$$W_a(\phi) = U(\psi_a(\phi)) - U(0) + kT [\log q_0(\phi) - \log q_0(\psi_a(\phi)) - \log J_a(\phi)]. \quad (17)$$

Change of variables gives

$$\mathbb{E}_{q_0} e^{-W_a/kT} = \int_{-\pi}^{\pi} q_0(y) e^{-[U(y)-U(0)]/kT} dy = Z_1/Z_0, \quad Z_0 = 1. \quad (18)$$

The generalized work includes the volume correction for the imposed escort; the map is not asserted to be a trajectory of unforced molecular dynamics.

All 1,536 single-point calculations succeed and their logs, gradients, atom identities and coordinates are independently replayed. The reference potential is explicitly the periodic cubic interpolant of 512 GFN2 values per rotor. Its 256-knot interpolation error at the additional midpoints is at most 10^{-8} eV in this study. Increasing the reference quadrature from 32,768 to 65,536 points changes the free energy by less than 10^{-10} eV. These numbers describe numerical consistency of this interpolated approximate potential, not electronic-structure accuracy.

Every molecule, escort, sample budget and repetition is retained. Total variation uses 64 equal angular bins; reference bin probabilities come from dense quadrature. Normalized finite-sample importance weights are biased estimates of the normalized target, while their unnormalized mean estimates its normalizer. The reported free energy is $-kT \log \hat{Z}$, with its finite-sample bias included in RMSE. At 2,048 draws, complete-work ESS ranges from 788 to 1,892 across the six nonidentity cases. Some

incomplete weights have higher ESS despite their biased target, demonstrating that ESS alone does not validate work accounting.

These controlled one-dimensional results validate the correction and its finite sample behavior on the specified targets. They do not establish unrestricted three-dimensional Boltzmann generation or an additional neural-training benefit of work weighting over the matched force-update control.